Exercise list R2b — Bond valuation, the term structure and the cost of debt

Answer key (full derivations in `lista-r2b-bond-valuation-solutions.pdf`)

Notation: $A(y,n) = [1-(1+y)^{-n}]/y$; price $P = C\,A(y,n) + F/(1+y)^n$ with per-period C and y.

1. Zero-coupon yields (Berk 6.4)

$YTM_n = (100/P_n)^{1/n} - 1$: e.g. $n=2$: $(100/91.05)^{1/2} - 1 = 1.098298^{1/2} - 1 = 4.80\%$. **(a)** 4.70%, 4.80%, 5.00%, 5.20%, 5.50% for 1–5 years. **(b)** Rising line through those points. **(c)** Upward sloping.

2. Holding-period return (Higgins 5.5)

(a) $HPR = [60 + (940 - 1{,}050)]/1{,}050 = -50/1{,}050 = -4.76\%$ (coupon $0.06 \times 1{,}000 = 60$). **(b)** Interest rates rose, or the issuer's perceived default risk rose; both raise the yield. Matches p. 410.

3. Seasoned semiannual bond (Ross 7.6)

20 half-years remain; $C = 34.5$, $y = 3.7\%$: $A(0.037, 20) = 13.958605$, $(1.037)^{-20} = 0.483532$. $P = 34.5(13.958605) + 1{,}000(0.483532) = 481.57 + 483.53 = \$965.10$ per \$1,000 face.

4. Callable vs. noncallable (Higgins 5.4)

$P^{\text{callable}} = P^{\text{straight}} - V^{\text{call}}$: lower price for the same promised flows, so the callable bond's YTM is **higher**; investors are paid for the option they sell to the issuer (exercised when rates fall).

5. Expected return vs. yield (Berk CT 8)

$1 + y = F/P$ uses promised payments; $1 + r_D = [F(1-p) + Cp]/P$ uses expected ones, which are lower when $p > 0$ and $C < F$, so $r_D < y$. Illustration $F = 100$, $P = 90$, $p = 5\%$, $C = 40$: $y = 11.11\%$, $r_D = 97/90 - 1 = 7.78\%$.

6. Credit spreads (Berk 6.24)

(a) $100/1.032 = 96.90\%$ of face. **(b)** $3.2 - 3.1 = 0.1\%$. **(c)** $4.9 - 3.1 = 1.8\%$. **(d)** Spreads widen as the rating falls (0.1, 1.1, 1.8%): higher default probability and lower recovery.

7. GMAC bond over time (Berk 6.16–6.18)

(16) $70(7.360087) + 1{,}000(0.558395) = 515.21 + 558.39 = \$1{,}073.60$. **(17)** $1{,}073.60 \times 1.06 = \$1{,}138.02$. **(18)** 9-year bond: $70(6.801692) + 1{,}000(0.591898) = \$1{,}068.02$ $(= 1{,}138.02 - 70)$.

8. Spot-rate pricing (Berk 6.10)

$P = 20/1.005 + 20/1.0055^2 + 1{,}020/1.0065^3 = 19.90 + 19.78 + 1{,}000.37 = \$1{,}040.05$ (YTM 1.295% APR).

9. Coupon for a new par issue (Ross 7.19)

Solve $930 = 40\,A(y, 40) + 1{,}000/(1+y)^{40}$: $y = 4.3736\%$ per half-year, APR 8.75% (EAR 8.94%). Set the new coupon at **8.75%** (not the old 8%).

10. HMK ratings (Berk 6.26)

(a) At 6.3%: $65(4.178207) + 1{,}000(0.736773) = \$1{,}008.36$. **(b)** $10{,}000{,}000/1{,}008.36 = 9{,}917.13 \to$ 9,917 bonds: \$9,917,000 face. **(c)** A (yield 6.5% = coupon). **(d)** At 7.5%: $65(4.045885) + 1{,}000(0.696559) = 959.54$, so BB: junk.

11. Interest-rate sensitivity (Berk 6.21–6.22)

(21) A 417.27 → 481.02: +15.28%; B 558.39 → 613.91: +9.94%; C 805.76 → 896.20: +11.23%; D 1,147.20 → 1,231.65: +7.36%. **(22)** Most sensitive A, least D; order A > C > B > D follows duration (15, 11.11, 10, 7.45 years): value received later is discounted more times.

12. Bonds M and N (Ross 7.31)

$r = 3.5\%$: $PV_1 = 1{,}100(12.094117)/1.511069 = 8{,}804.05$; $PV_2 = 1{,}400(9.663334)/2.620172 = 5{,}163.27$; $PV_F = 20{,}000/3.959260 = 5{,}051.45$. $P_M = \$19{,}018.78$; $P_N = \$5{,}051.45$.

13. Bootstrapping (G&T 10A.3)

$d_t = (100 - c_t \sum_{k<t} d_k)/(100 + c_t)$: $d = 0.956938,\ 0.906812,\ 0.857153,\ 0.814397$. Spot yields 4.50%, 5.01%, 5.27%, 5.27% (5.2723%, then 5.2667%). Annuity prices 0.956938, 1.863750, 2.720903, 3.535300; annuity yields 4.50%, 4.84%, 5.05%, 5.13%.

14. Implied call value (Ross 7.32)

Weights $w = 3.75/5.5 = 0.681818$ on the 6.5% bond, 0.318182 on the 12% bond give an 8.25% coupon. Ask: $0.681818(106.375) + 0.318182(134.96875) = 115.4730$; call $= 115.4730 - 103.5 = 11.9730\%$ of face $\approx$ **\$119.73** per \$1,000 (bid prices: \$119.33).

15. Defaultable bond on a tree (G&T 7.3)

$\pi_0 = 14/30 = 0.466667$, $\pi_U = 14.8/190 = 0.077895$, $\pi_D = 43.6/60 = 0.726667$; $V_U = 105/1.04 = 100.9615$, $V_D = 89.9667/1.04 = 86.5064$; $V_0 = 5/1.04 + 93.2521/1.04 = 4.8077 + 89.6655 = \$94.47$. Promised yield 8.10% per half-year vs. risk-neutral expected return 4%; gap to the default-free twin $101.89 - 94.47 = 7.4129 = 55(0.145778)/1.0816$.